Lab 4 Conceptual Design

Business Requirements

A university library needs a new dbms. The system needs to handle users borrowing and returning books and keep track of the overall status of the library. The library is located across several sites and all students can borrow up to 10 books at a time from any of the sites. For some popular books, the library keeps multiple copies across different sites to facilitate students' access. Details of site manager, authors and publishers are also kept.

Extra requirements:

- Each site has one manager and managers can only manage one site.
- Assume books only have one author.

Step 1: Identify Entities

Step 2: Find Relationships

With reference to the business requirements, complete the relationship matrix below:

Step 3: Draw Rough ER Diagram

Step 4: Fill in Cardinality

Step 5: Define Primary Keys

Step 6: Draw Key-Based ERD

This step involves:

- ***Eliminating*** *many-to-many relationships*, and
- ***Adding*** *Primary* and *Foreign keys* for each *entity* (started in the last step)

Step 7: Identify Attributes

Step 8: Map Attributes

Step 9: Draw Fully Attributed E-R Diagram

Step 10: Check ER Diagram Results

Look at your diagram from the point of view of a system owner or user. Is everything clear? Check through the Cardinality pairs. Also, look over the list of attributes associated with each entity to see if anything has been omitted. Write down any assumptions you may have made along the way.

Questions and further improvements

1. What is the aim of this system?
    a. Do we need information about staff?

2. What about books written by more than one author?

3. What if we want to add functionality? For example, to find out if a book is available.

    What changes do we need to consider?
    a. Extra entities?
    b. Extra attributes?